

Given (for all proofs below)

$V = IR + E_{mf}$ by Ohm's Law with back EMF for a DC motor

$E_{mf} = \omega\kappa_e$ by the Back EMF equation of a motor

$\tau = I\kappa_t$ by the Motor torque equation

$\tau = J\alpha + B\omega + \tau_{load} \text{sign}(\omega)$ by the Rotational equation of motion

The value of ω and α (Only for the proof directly below this)

Find V voltage

$V = IR + E_{mf}$, $E_{mf} = \omega\kappa_e$, $\tau = I\kappa_t$, $\tau = J\alpha + B\omega + \tau_{load} \text{sign}(\omega)$, and the value of ω and α because they are given

$I\kappa_t = J\alpha + B\omega + \tau_{load} \text{sign}(\omega)$ By substitution

$I = \frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t}$ By Division

$-IR = E_{mf} - V$ By Subtraction

$I = \frac{E_{mf} - V}{-R}$ By Division

$\frac{E_{mf} - V}{-R} = \frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t}$ By substitution

$\frac{\omega\kappa_e - V}{-R} = \frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t}$ By substitution

$\omega\kappa_e - V = \frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t}(-R)$ By multiplication

$-V = \frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t}(-R) - \omega\kappa_e$ By subtraction

$V = -\left(\frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t}(-R) - \omega\kappa_e\right)$ By division

$V = \frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t}(R) + \omega\kappa_e$ By substitution

$V = \frac{J\alpha R}{\kappa_t} + \frac{B\omega R}{\kappa_t} + \frac{\tau_{load} \text{sign}(\omega) R}{\kappa_t} + \frac{\omega\kappa_e \kappa_t}{\kappa_t}$ By substitution

$V = \frac{J\alpha R}{\kappa_t} + \frac{B\omega R + \omega\kappa_e \kappa_t}{\kappa_t} + \frac{\tau_{load} \text{sign}(\omega) R}{\kappa_t}$ By substitution

$V = \frac{J\alpha R}{\kappa_t} + \frac{\omega(BR + \kappa_e \kappa_t)}{\kappa_t} + \frac{\tau_{load} \text{sign}(\omega) R}{\kappa_t}$ By substitution

Define K_v by $K_v = \frac{BR + \kappa_e \kappa_t}{\kappa_t}$

Define K_a by $K_a = \frac{JR}{\kappa_t}$

Define K_s by $K_s = \frac{\tau_{load} R}{\kappa_t}$

$V = K_a\alpha + K_v\omega + K_s \text{sign}(\omega)$

Define a function V by $V(\omega, \alpha) = K_a\alpha + K_v\omega + K_s \text{sign}(\omega)$

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Given (Only for the proof directly below this)

The value of ω

Find the maximum α angular acceleration at some angular velocity ω

$V = IR + E_{mf}$, $E_{mf} = \omega\kappa_e$, $\tau = I\kappa_t$, $\tau = J\alpha + B\omega + \tau_{load} \text{sign}(\omega)$, and the value of ω because they are given

Note: The VEX motor controller has current limits so we must also account for those.

Define $I_{\min}$ and $I_{\max}$ as the current limits and define I_{clamp} as the clamped value of I

Define $V_{\max}$ as the maximum voltage and set it to V when finding the max acceleration

Define $\alpha_{\max}$ as the maximum voltage and set it to α when finding the max acceleration

$I_{\text{clamp}} = \min(\max(I, I_{\min}), I_{\max})$ by the definition of clamping

$I_{\text{clamp}} = \min\left(\max\left(\left(\frac{E_{\text{mf}} - V_{\max}}{-R}\right), I_{\min}\right), I_{\max}\right)$ by substitution

$I_{\text{clamp}} = \min\left(\max\left(\left(\frac{\omega \kappa_e - V_{\max}}{-R}\right), I_{\min}\right), I_{\max}\right)$ by substitution

Define $V_{\max} = V$ as the max voltage

$I_{\text{clamp}} = \min\left(\max\left(\left(\frac{\omega \kappa_e - V_{\max}}{-R}\right), I_{\min}\right), I_{\max}\right)$ by substitution

Restrict I by $I_{\text{clamp}} = I$ by the note

$-J\alpha_{\max} = B\omega + \tau_{\text{load}} \text{sign}(\omega) - \tau$ by Subtraction

$\alpha_{\max} = \frac{B\omega + \tau_{\text{load}} \text{sign}(\omega) - \tau}{-J}$ by Division

$\alpha_{\max} = \frac{B\omega + \tau_{\text{load}} \text{sign}(\omega) - I\kappa_t}{-J}$ by Substitution

$\alpha_{\max} = \frac{B\omega + \tau_{\text{load}} \text{sign}(\omega) - I_{\text{clamp}}\kappa_t}{-J}$ by Substitution

Define a function α by $\alpha_{\max}(\omega) = \frac{B\omega + \tau_{\text{load}} \text{sign}(\omega) - I_{\text{clamp}}\kappa_t}{-J}$

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Find $\omega(t)$ given that the motor is constantly accelerating

$\alpha_{\max}(\omega) = \frac{V_{\max} - K_v\omega - \text{sgn}(\omega)K_s}{K_a}$

Let $s = \text{sgn}(\omega) \in \{1, -1\}$

$\frac{d\omega}{dt} = \frac{V_{\max} - K_v\omega - sK_s}{K_a}$

$\frac{d\omega}{dt} + \frac{K_v}{K_a}\omega = \frac{V_{\max} - sK_s}{K_a}$

$I(t) = e^{\int \frac{K_v}{K_a} dt}$

$I(t) = e^{\frac{K_v}{K_a} t}$

$\omega(t) = \frac{1}{I(t)} \left(\int_0^t I(\tau) Q d\tau + I(0)\omega_0 \right)$

$\omega(t) = \frac{1}{e^{\frac{K_v}{K_a} t}} \left(\int_0^t e^{\frac{K_v}{K_a} \tau} \frac{V_{\max} - sK_s}{K_a} d\tau + \omega_0 \right)$

$\omega(t) = \frac{1}{e^{\frac{K_v}{K_a} t}} \left(\frac{V_{\max} - sK_s}{K_a} \frac{K_a}{K_v} \left(e^{\frac{K_v}{K_a} \tau} - 1 \right) + \omega_0 \right)$

$\omega(t) = \frac{V_{\max} - sK_s}{K_v} \frac{e^{\frac{K_v}{K_a} \tau} - 1}{e^{\frac{K_v}{K_a} t}} + \omega_0 \left(e^{-\frac{K_v}{K_a} t} \right)$

$\omega(t) = \frac{V_{\max} - sK_s}{K_v} + \left(\omega_0 - \frac{V_{\max} - sK_s}{K_v} \right) e^{-\frac{K_v}{K_a} t}$

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Find a continuous state-space model in the form

$\dot{x} = Ax + Bu$

$y = Cx + Du$

$v = K_a\alpha + K_v\omega + K_s \text{sign}(\omega)$ because it is given

let $u = v$ and $x = \begin{pmatrix} \omega \\ \text{sign}(\omega) \end{pmatrix}$

$\alpha = \dot{\omega}$ by the definition of acceleration

$\dot{x}(1 \ 0) = \dot{\omega}$ by the derivative

$\alpha = \dot{x}(1 \ 0)$ by substitution

$\text{sign}(\omega) = x(0 \ 1)$

$u = K_a \alpha + K_v \omega + K_s \text{sign}(\omega)$ by substitution

$K_a \dot{x}(1 \ 0) = u - K_v x(1 \ 0) - K_s x(0 \ 1)$ by subtraction

$K_a \dot{x}(1 \ 0) = (-K_v \ -K_s)x + u$ by substitution

$\dot{x}(1 \ 0) = \left(-\frac{K_v}{K_a} \ -\frac{K_s}{K_a}\right)x + \frac{1}{K_a}u$ by division

$\dot{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \left(-\frac{K_v}{K_a} \ -\frac{K_s}{K_a}\right)x + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \frac{1}{K_a}u$ by multiplication

$\dot{x} = \begin{pmatrix} -\frac{K_v}{K_a} & -\frac{K_s}{K_a} \\ 0 & 0 \end{pmatrix}x + \begin{pmatrix} \frac{1}{K_a} \\ 0 \end{pmatrix}u$ by substitution

Let $A = \begin{pmatrix} -\frac{K_v}{K_a} & -\frac{K_s}{K_a} \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} \frac{1}{K_a} \\ 0 \end{pmatrix}$

Let $y = (\omega)$

$y = x(1 \ 0)$ by substitution

Let $C = (1 \ 0)$ and $D = 0$

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Given

$$A_d = e^{A_c T}$$

$$B_d = \int_0^T e^{A_c \tau} d\tau B_c$$

$$C_d = C_c$$

$$D_d = D_c$$

$$A_c = \begin{pmatrix} -\frac{K_v}{K_a} & -\frac{K_s}{K_a} \\ 0 & 0 \end{pmatrix}$$

$$B_c = \begin{pmatrix} \frac{1}{K_a} \\ 0 \end{pmatrix}$$

$$C_c = (1 \ 0)$$

$$D_c = 0$$

Find a discrete state-space model in the form

$$x_{k+1} = A x_k + B u_k$$

$$y_k = C x_k + D u_k$$

$$A_d = e^{\begin{pmatrix} -\frac{K_v}{K_a} & -\frac{K_s}{K_a} \\ 0 & 0 \end{pmatrix} T} \text{ by substitution}$$

$$A_d = e^{\begin{pmatrix} -T \frac{K_v}{K_a} & -T \frac{K_s}{K_a} \\ 0 & 0 \end{pmatrix}} \text{ by substitution}$$

$$\text{Let } a = -T \frac{K_v}{K_a} \text{ and } b = -T \frac{K_s}{K_a}$$

$$A_d = e^{\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}} \text{ by substitution}$$

$$\mathbf{A}_d = \begin{pmatrix} e^a & b \frac{e^a - e^0}{a-0} \\ 0 & e^0 \end{pmatrix} \text{ by the exponential of an upper-triangular matrix formula}$$

$$\mathbf{A}_d = \begin{pmatrix} e^{-T \frac{K_v}{K_a}} & -T \frac{K_s}{K_a} e^{-T \frac{K_v}{K_a}} \\ 0 & 1 \end{pmatrix} \text{ by substitution}$$

$$\mathbf{A}_d = \begin{pmatrix} e^{-T \frac{K_v}{K_a}} & K_s e^{-T \frac{K_v}{K_a}} \\ 0 & 1 \end{pmatrix} \text{ by substitution}$$

$$\mathbf{B}_d = \int_0^T \begin{pmatrix} e^{-\tau \frac{K_v}{K_a}} & K_s e^{-\tau \frac{K_v}{K_a}} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{K_a} \\ 0 \end{pmatrix} d\tau \text{ by substitution}$$

$$\mathbf{B}_d = \int_0^T \begin{pmatrix} e^{-\tau \frac{K_v}{K_a}} \\ 0 \end{pmatrix} d\tau \text{ by substitution}$$

$$\mathbf{B}_d = \begin{pmatrix} \frac{1}{K_a} \int_0^T e^{-\tau \frac{K_v}{K_a}} d\tau \\ 0 \end{pmatrix} \text{ by substitution}$$

$$\text{Let } I_d = \int_0^T e^{-\tau \frac{K_v}{K_a}} d\tau$$

$$\mathbf{B}_d = \begin{pmatrix} I_d \\ 0 \end{pmatrix} \text{ by substitution}$$

$$I_d = e^{-0 \frac{K_v}{K_a}} - e^{-T \frac{K_v}{K_a}} \text{ by FTC}$$

$$I_d = 1 - e^{-T \frac{K_v}{K_a}} \text{ by substitution}$$

$$\mathbf{B}_d = \begin{pmatrix} \frac{1 - e^{-T \frac{K_v}{K_a}}}{K_a} \\ 0 \end{pmatrix} \text{ by substitution}$$

$$\mathbf{C}_d = (1 \ 0) \text{ by substitution}$$

$$\mathbf{D}_d = 0 \text{ by substitution}$$

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Given

$$u_k^* = \arg \min_{u_k} \Sigma$$

Find K based on $\mathbf{Q}$ and $\mathbf{R}$ in LQR

$$\mathbf{A}^T \mathbf{S} \mathbf{A} - \mathbf{S} - \mathbf{A}^T \mathbf{S} \mathbf{B} (\mathbf{R} + \mathbf{B}^T \mathbf{S} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{S} \mathbf{A} + \mathbf{Q} = 0 \text{ by the discrete algebraic Riccati equation}$$

$$\mathbf{K} = (\mathbf{R} + \mathbf{B}^T \mathbf{S} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{S} \mathbf{A} \text{ by the discrete LQR result solved by the discrete algebraic Riccati equation}$$

$$\mathbf{A}^T \mathbf{S} \mathbf{A} - \mathbf{S} - \mathbf{A}^T \mathbf{S} \mathbf{B} (\mathbf{R} + \mathbf{B}^T \mathbf{S} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{S} \mathbf{A} + \mathbf{Q} = 0$$

$$\mathbf{K} = (\mathbf{R} + \mathbf{B}^T \mathbf{S} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{S} \mathbf{A}$$

$$\mathbf{A} = \begin{pmatrix} e^{-T \frac{K_v}{K_a}} & K_s e^{-T \frac{K_v}{K_a}} \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{B} = \begin{pmatrix} \frac{1 - e^{-T \frac{K_v}{K_a}}}{K_a} \\ 0 \end{pmatrix}$$

$$\mathbf{C} = (1 \ 0)$$

$$\mathbf{D} = 0 \text{ by substitution}$$